Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

Abstract

CCS Concepts

• **Human-centered computing → Mixed / augmented reality**; • **Computing methodologies → Rendering**; *Computer vision.*

Keywords

eye tracking, deep learning, computer graphics

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1 Introduction

2 Related works

To compare our work with existing eye convergence and accommodation tracking, we review related works from two main areas: physical feature-based experimental designs and virtual eye modeling and neural rendering. To clearly show the technical features of previous works, Table 1 categorizes and compares these methods based on their hardware setups, algorithms, operating modes, and tracking accuracy.

2.1 Eye Accommodation Tracking

In stereoscopic and near-eye displays, the Vergence-Accommodation Conflict (VAC) [Hoffman et al. 2008]arises when the eyes converge to a virtual 3D depth while the crystalline lens remains focused on the physical display plane. This mismatch can reduce fusion performance and cause visual discomfort . Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. However, vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems. Higher-order Purkinje images provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al.[2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al.[2023] extend this direction by collecting multi-subject Purkinje measurements and combining a Purkinje imaging setup with a **Multi-Layer Perceptron (MLP)**-based regression model to estimate dynamic vergence and accommodation across users.

These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye geometry. Motivated by this limitation, we first build a controllable 3D optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

2.2 Computational Eye Modeling

Computational eye models are widely used to generate labeled data for gaze estimation. UnityEyes, NVGaze, and RIT-Eyes synthesize controllable variations in gaze, ocular geometry, appearance, and illumination for training eye-tracking models [Kim et al. 2019; Nair et al. 2020; Wood et al. 2016]. These systems capture visible eye-region cues effectively, but they primarily target eye rotation, pupil localization, and segmentation rather than accommodation-dependent changes within the crystalline lens.

Neural eye representations improve visual realism and relighting. EyeNeRF combines an explicit eyeball surface with implicit volumetric representations for view synthesis, gaze animation, corneal refraction, and relighting [Li et al. 2022]. Khademi et al. extend a similar hybrid formulation to near-field point sources and structured illumination [Khademi et al. 2026]. However, these methods focus on photorealistic appearance and do not explicitly predict P3/P4 formation or accommodation-dependent internal ocular geometry.

Purkinje-based systems model internal reflections more directly. Wu et al., Idrizovi'c et al., and OpenIrisDPI use P1 and P4 for accurate eye-rotation tracking [Idrizović et al. 2025; Ressmeyer et al. 2026; Wu et al. 2023]. Lu et al. further model higher-order reflections with multiple cameras and infrared LEDs to estimate vergence and accommodation [Lu et al. 2020]. Ozhan et al. simulate P1–P4 across accommodation and vergence states, validate the predictions against real captures, and use the resulting features for dynamic estimation [Ozhan et al. 2023]. Our work follows this direction by using a controllable optical eye model to predict how accommodation, vergence, and LED–camera geometry affect P1–P4 formation.

2.3 Physics-Based Optical Design

Differentiable optical design provides a possible framework for optimizing the illumination and imaging layout of a Purkinje capture system. Na et al. show that image-domain losses can be propagated through differentiable ray tracing to physical optical parameters [Na et al. 2024]. For our system, the same principle could be applied to LED position, camera pose, focal distance, aperture, and field of view.

The optimization criteria should follow the capture behavior of higher-order Purkinje reflections. Lu et al. show that P3 is weaker than P1, lies on a different focal plane, and may leave the camera

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Table 1: Comparison of different eye and accommodation tracking methodologies. Accuracy is qualitatively categorized into High (High), Mid (Mid), and Low (Low) according to the reported performance for the corresponding tracking task.

No.	Ref.	Camera	Illum.	Method	Algorithm	Target	Mode	Accuracy
1	[Cornsweet and Crane 1973]	Multiple	IR	DPI	Geometric	Vergence	Desktop	Mid
2	[Morimoto et al. 2002]	Single	IR	PCCR	Geometric	Vergence	Desktop	Mid
3	[Sturgeon et al. 2025]	Multiple	IR	PCCR	Geometric	Vergence	Near-eye	Mid
4	[Bontha and Arefin 2024]	Multiple	IR	Mocap	Geometric	Vergence	Near-eye	Low
5	[Lu et al. 2020]	Multiple	IR	Purkinje	Learning	Both	Near-eye	High
6	[Ozhan et al. 2023]	Single	IR	Purkinje	Hybrid	Both	Desktop	High
7	[Nair et al. 2020]	Single	None	Ray Tracing	Rendering	Vergence	Synthetic	Mid
8	[Li et al. 2022]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low
9	[Khademi et al. 2026]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low
10	[?]	Single	IR	DPI	Regression	Depth	Desktop	Mid
11	[?]	Single	IR	Purkinje	Geometric	Cornea	Synthetic	Low
12	[?]	Single	IR	DPI	Geometric	Vergence	Desktop	High
13	[?]	Multiple	IR	NeRF	Learning	Gaze	Synthetic	Mid

field of view as gaze changes; their system therefore uses multiple cameras and infrared LEDs [Lu et al. 2020]. Ozhan et al. further show that the camera–LED–eye geometry affects the observed P3/P4 locations and configure the system to improve reflection separation while avoiding vignetting [Ozhan et al. 2023]. These results suggest optimizing P3/P4 visibility, sharpness, field-of-view coverage, and separation from other reflections.

The selected layout must also remain practical to assemble, calibrate, and evaluate at the sensor level. Prior work on replicating a controllable AR haploscope highlights the implementation challenges of physical optical platforms [Bontha and Arefin 2024] and Wu et al. show that reliable P1/P4 measurement depends on sufficient image resolution and accurate reflection localization [Wu et al. 2023]. These considerations motivate evaluating optical layout, calibration, and sensor-level detectability together.

The present work does not perform end-to-end differentiable optimization. Instead, it uses geometric-optics simulation to compare candidate LED–camera configurations and to predict whether P1–P4 remain visible, sharp, and separable under realizable capture conditions. Differentiable layout optimization and synthetic-to-real adaptation are left as future extensions.

3 Optical Background

3.1 Challenges of reconstructing Purkinje reflections via 3D Gaussian Splatting (3DGS).

Standard 3DGS struggles to consistently preserve Purkinje reflections because its reconstruction formulation does not fully match their physical image-formation process. 3DGS represents view-dependent appearance by using low-order Spherical Harmonics (SH) [Kerbl et al. 2023], but accurately modeling high-frequency, view-dependent specular appearance remains challenging [**??**]. More importantly, P3 and P4 are formed through transmission and reflection across the multiple ocular interfaces of eyes, whereas rasterization-based 3DGS does not explicitly model these secondary optical paths [**?**]. In addition, Lu et al. [Lu et al. 2020] and Wu et al. [Cornsweet and Crane 1973] show that P3 is formed at a different optical focal plane from P1 and P4, while higher-order Purkinje reflections are relatively weak and therefore more susceptible to defocus and low contrast. Finally, Purkinje locations vary jointly with accommodation, vergence, subject-specific ocular geometry, and camera–illumination configuration [Lu et al. 2020; Ozhan et al. 2023]. Purkinje reflections are better characterized as dynamic optical signals governed by ocular optics rather than stable surface textures, making it difficult for standard 3DGS to reconstruct them consistently with their underlying physical behavior.

3.2 Capturing Purkinje image

Dual Purkinje Image (DPI) tracking can provide higher precision than conventional pupil–corneal-reflection tracking by using the relative motion of P1 and P4 rather than the pupil center [Cornsweet and Crane 1973; Ressmeyer et al. 2026]. However, obtaining reliable Purkinje features from real eye images remains challenging. Wu et al. [Cornsweet and Crane 1973] show that P1 and P4 can be imaged together while P3 is out of focus and P2 largely overlaps with P1; similarly, Ahn et al. [**?**] report that P4 is substantially smaller and darker than P1, making its detection less reliable. Therefore, the existence of a valid reflection path does not necessarily guarantee a usable image feature, as its visibility further depends on optical focus, reflection strength, occlusion, and imaging noise [Cornsweet and Crane 1973; **?**]. Camera and illumination configuration also affect whether these reflections remain observable across different eye orientations, making Purkinje detectability a joint consequence of ocular geometry and the imaging configuration [Cornsweet and Crane 1973; Lu et al. 2020]. Nevertheless, Purkinje geometry carries useful information about the underlying ocular surfaces, with previous phakometric studies showing that the position and geometry of these reflections can be used to infer ocular surface curvature and alignment [**?**].

3.3 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since

P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [**???**]. Existing Purkinje imaging systems also provide practical guidance for hardware geometry. Both Lu et al. and Ozhan et al. place infrared LEDs close to the camera optical axis to maximize Purkinje visibility while keeping the reflections inside the camera field of view. At the same time, camera positions must balance the visibility of higher-order Purkinje images against eyelid and eyelash occlusions, glint overlap, and sensor saturation. These observations suggest that LED and camera placement is not arbitrary but jointly determines the brightness, separation, and detectability of P1–P4. Once the capture geometry is determined, synthetic rendering provides an efficient way to evaluate different configurations before physical experiments. RIT-Eyes demonstrates that physically based rendering can generate realistic near-eye images under controllable illumination, while EyeNeRF and subsequent hybrid eye representations further show that explicit corneal geometry is necessary for reproducing specular reflections and refraction, whereas implicit neural representations are better suited for modeling complex periocular appearance and relighting. These observations motivate the use of explicit optical eye models for Purkinje simulation before subsequent NeRF or 3D Gaussian reconstruction, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023]

4 Method

4.1 3D Gaussian Splatting

3D Gaussian Splatting is an explicit 3D scene representation and differentiable rendering method designed for novel view synthesis. The core idea is modeling the geometric structure and viewpoint-dependent appearance of a scene in 3D space using a set of optimizable Gaussian primitives expressed as

$$\mathcal{G}_i = \left\{ \boldsymbol{\mu}_i,\ \Sigma_i,\ \alpha_i,\ \mathbf{c}_i \right\}, \tag{1}$$

Here, $\boldsymbol{\mu}_i$ denotes the 3D position of the i-th Gaussian, $\mathbf{s}_i$ its scale, $\mathbf{R}_i$ its rotation, α_i its opacity, and $\mathbf{c}_i$ the spherical harmonics coefficients used to represent view-dependent appearance.

The input to 3DGS consists of a set of multi-view images capturing the same scene from different viewpoints, structure-from-Motion (SfM) estimates the camera parameters and reconstructs a sparse 3D point cloud, which is subsequently used to initialize the Gaussian primitives. [Kerbl et al. 2023; **?**]. The scale and rotation jointly define the covariance matrix as

$$\Sigma_i = \mathbf{R}_i \mathbf{S}_i \mathbf{S}_i^T \mathbf{R}_i^T, \tag{2}$$

where $\mathbf{S}_i$ is a diagonal scaling matrix constructed from $\mathbf{s}_i$, and $\mathbf{R}_i$ determines the orientation of the anisotropic Gaussian.

Once the Gaussian primitives are initialized and parameterized, they are projected onto the image plane, ordered according to depth, and accumulated using front-to-back alpha compositing to generate the rendered image. Standard 3DGS combines a pixel-wise L_1 loss with a D-SSIM loss [Kerbl et al. 2023]:

$$\mathcal{L}_{\mathrm{GS}} = (1-\lambda)\mathcal{L}_1 + \lambda\mathcal{L}_{\text{D-SSIM}}, \tag{3}$$

where $\mathcal{L}_1$ measures the pixel-wise photometric difference between the rendered and ground-truth images, $\mathcal{L}_{\text{D-SSIM}}$ measures their local structural dissimilarity, and λ controls the relative contribution of the two terms. The reconstruction loss is back-propagated to jointly optimize the Gaussian positions, scales, rotations, opacities, and SH coefficients.

In parallel with parameter optimization, 3DGS performs adaptive density control to dynamically refine the Gaussian population. Gaussians with large positional gradients are cloned when they are sufficiently small and split when they have a large spatial extent, introducing additional primitives in under-reconstructed regions. Conversely, Gaussians with very low opacity or excessively large spatial or screen-space extent are periodically removed. By alternating differentiable rendering, gradient-based optimization, and adaptive density control, the initially sparse Gaussian representation progressively evolves into a dense scene representation capable of high-quality novel-view rendering [Kerbl et al. 2023].

5 Results and Discussions

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